

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

**Interconnection of Large Loads to the Interstate Transmission
System**

) **Docket No.**
) **RM26-4-000**

Initial Comments of L. Lynne Kiesling, Ph.D.

November 21, 2025

Executive Summary

This comment evaluates the Department of Energy's Advance Notice of Proposed Rulemaking (ANOPR) regarding large-load interconnection through the lens of transaction-cost economics, regulatory economics, and technological change. The rapid growth of AI data center load presents a critical coordination challenge due to the temporal mismatch between rapid data center deployment (18–24 months) and slow grid infrastructure expansion (7–10 years).

To address this pacing mismatch and ensure nondiscriminatory access, the Commission should adopt a modular, principles-based framework grounded in three economic propositions:

- **Flexibility as a System Value:** Modern data centers possess significant latent flexibility through workload shifting, UPS-based frequency response, and hybrid on-site generation. Current interconnection rules treat these loads as static and inflexible, ignoring capabilities that could significantly reduce system stress and upgrade costs.
- **Reducing Transaction Costs:** Legacy interconnection rules create high asset specificity and hold-up risks, driving developers toward inefficient bypass strategies. Co-location and hybrid configurations are rational economic responses to these high transaction costs ("make vs. buy" decisions) and should be facilitated rather than penalized by extending generator-style interconnection regimes to loads.
- **Foundational Cost Allocation Reform:** The ANOPR's suggestion of 100% participant funding for upgrades relies on flawed embedded-cost methodologies that ignore marginal impacts. Efficient cost recovery should instead follow Ramsey-Boiteux principles, allocating fixed costs inversely to elasticity. Because large loads are highly elastic and can relocate or self-supply, assigning them full upgrade costs encourages inefficient grid defection.

Recommendations: The Commission should establish a federal framework that lowers barriers to flexible resources, modernizes market participation rules for technologies like thermal energy storage (TES), and reforms cost allocation to reflect marginal impacts. This approach will encourage the integration of grid-supportive flexibility, reducing the need for costly infrastructure while ensuring reliability and competitive neutrality.

I. Introduction and Interest of the Commenter

The accelerating growth of data center and artificial intelligence (AI) electricity demand is reshaping the U.S. power system. Forecasts suggest that U.S. data center load, roughly 25 gigawatts (GW) in 2024, may exceed 80 GW by 2030. Annual electricity consumption attributable to these facilities could rise by approximately 400 terawatt-hours (TWh) over the decade.¹ Globally, the International Energy Agency projects data center electricity use to more than double, increasing from an estimated 415 TWh in 2024 to roughly 945 TWh by 2030.² These estimates remain uncertain because of rapid evolution in AI chip efficiency, model architectures, inference patterns, cooling requirements, and data center design. Several recent analyses describe these rapid forecast revisions as “forecast whiplash,” reflecting the difficulty of relying on planning institutions built for a world of slow, predictable demand growth.³

I am the Director of the Institute for Regulatory Law & Economics at Northwestern University, as well as a Nonresident Senior Fellow at the American Enterprise Institute and a member of the External Faculty of the Santa Fe Institute; my comments represent my own analysis and do not reflect the opinions of any of those organizations. I submit this comment as an economist whose research sits at the intersection of electricity market design, utility regulation, institutional economics, and technological change. My published work examines how modern grid technologies interact with legacy regulatory structures, how regulatory pacing failures impede innovation, and how transaction costs shape investment decisions. I also write translational research on current issues in utility regulation and technological change, including data center energy use. My comments rely on a selection of my relevant publications, including:

- Lee, Yoon-Ho Alex Lee and L. Lynne Kiesling. 2026. “[Epistemic Humility and Adaptive Regulation](#).” *New York University Journal of Law and Liberty*, forthcoming.

¹ Energy Information Administration, “[Annual Energy Outlook 2025](#)”; Alastair Green, Humayun Tai, Jesse Noffsinger, Pankaj Sachdeva, Arjita Bhan, and Raman Sharma, “[How Data Centers and the Energy Sector Can Sate AI’s Hunger for Power](#),” McKinsey & Company, 2024; OnLocation Incorporated, “[AEO 2025 Trends: Growing Electricity Demand from Data Centers](#),” 2025. All accessed November 21, 2025.

² International Energy Agency. 2025. “[Energy and AI](#),” 2025. Accessed November 21, 2025.

³ Ian Goldsmith and Zach Byrum, “[Powering the US Data Center Boom: Why Forecasting Can Be So Tricky](#),” World Resources Institute, September 17, 2025. Accessed November 21, 2025.

- [Microgrids, Nukes, and Novel Tariffs: How Hyperscale Data Centers are Redefining the Electricity Make-or-Buy Decision](#), Knowledge Problem, 2025.
- [Data Center Energy Use Series I-V](#), Knowledge Problem, 2024.
- Kiesling, L. Lynne. 2024. “[The Pacing Problem: Using the IAD Framework to Model Technological and Institutional Change in a Regulated Industry](#).” *Journal of Political Institutions and Political Economy* 5(4): 523-554.
- Mays, Jacob, Michael T. Craig, L. Lynne Kiesling, Joshua Macey, Blake Shaffer, and Han Shu. 2022. “[Private Risk and Social Resilience in Liberalized Electricity Markets](#).” *Joule*.
- Kiesling, Lynne. 2021. [The Essential Ronald Coase](#). Vancouver: Fraser Institute.
- Kiesling, Lynne. 2016. “[Implications of Smart Grid Innovation for Organizational Models in Electricity Distribution](#),” in Chen-Ching Liu, Stephen McArthur, and Seung-Jae Lee, eds., *Wiley Smart Grid Handbook*. London: Wiley.
- Kiesling, Lynne, and Andrew Kleit, editors. 2009. [Electricity Restructuring: The Texas Story](#). AEI Press.
- Kiesling, Lynne. 2009. [Deregulation, Innovation, and Market Liberalization: Electricity Restructuring in a Constantly Evolving Environment](#). Routledge Studies in Business Organizations and Networks. London: Routledge.

This comment evaluates the Department of Energy’s ANOPR on large-load interconnection through three related economic propositions. First, flexibility has become the highest-value attribute in modern power systems, largely because it a novel, technology-enabled manifestation of the price elasticity that power systems have lacked historically. Second, legacy interconnection rules create substantial market barriers for new technologies—particularly digital demand controls, hybrid on-site generation, flexible and hyper flexible loads, and microgrid-ready designs—that could otherwise mitigate system stress. Third, cost allocation reform is foundational: embedded-cost and fully distributed cost (FDC) methodologies conflict with marginal-cost principles, Ramsey-Boiteux efficiency conditions, and Bonbright’s criteria for sound regulation. Together, these three propositions support a federal approach that lowers system-wide transaction costs and encourages flexible technologies and business models through modular and principles-based methods.

II. Clarifying the Problem the ANOPR Is Trying to Solve

The ANOPR responds to a sharp shift in electricity demand dynamics driven by the rapid expansion of AI-centric data centers. After more than a decade of flat or marginal load growth—from 2008 to 2021, U.S. electricity demand rose by only 0.1 percent annually—forecasts have shifted dramatically upward.⁴ This reversal reflects the rise of a new

⁴ John D. Wilson, Sophie Meyer, Zach Zimmerman, and Rob Gramlich, [Power Demand Forecasts Revised Up for Third Year Running, Led By Data Centers](#), Grid Strategies, November 2025. Accessed November 21, 2025.

class of large, concentrated, and time-sensitive loads whose operational profiles differ substantially from historic industrial and commercial consumption.

1. Large-load demand and the intertemporal coordination challenge

Data centers can be built in eighteen to twenty-four months, enabled by modular construction and rapid IT deployment cycles.⁵ In contrast, major power system infrastructure upgrades often require seven to ten years to site, permit, and build. This temporal mismatch forces utilities and developers to make high-stakes decisions under deep uncertainty about future load trajectories.

The core policy challenge is therefore not simply the magnitude of new demand. It is a coordination problem under uncertainty in which actors with heterogeneous expectations and incomplete information must align investment, infrastructure, and operational decisions across very different time scales.

2. Cross-border impacts and interstate relevance

Because many hyperscale facilities seek to interconnect directly to the high-voltage transmission system, their effects propagate across regional networks. A single campus requesting 500 to 1,000 MW, comparable to the load of a mid-sized city, can influence congestion patterns, thermal limits, and power flows across an entire regional transmission organization (RTO) footprint.

Multiple authorities have highlighted these interstate implications. NERC has identified new data centers as a primary driver of rising peak-load forecasts and as an emerging reliability concern.⁶ PJM and other RTOs have documented significant upward revisions to multi-year load forecasts driven largely by data center development, alongside billions of dollars in transmission expansion required to serve these loads.

Because the impacts spill across state boundaries, allocating upgrade costs has already become contentious. This interstate character creates a legitimate federal interest in ensuring that processes governing transmission access for large loads are nondiscriminatory and account properly for cross-border consequences.

3. Current institutional fragmentation

Large-load interconnection and cost recovery are now shaped by a patchwork of state policies and utility practices. States such as Ohio, Michigan, and Georgia have adopted large-load tariffs and new cost-recovery mechanisms designed to shield retail ratepayers from the risks of large, uncertain industrial loads. Several counties in Georgia and Virginia have enacted restrictive zoning rules or moratoria on data center development, complicating siting decisions.

⁵ Jones Lang LaSalle. [“2025 Global Data Center Outlook,”](#) 2025. Accessed November 21, 2025.

⁶ North American Electric Reliability Corporation, [2025 State of Reliability: Overview](#), June 2025. Accessed November 21, 2025.

Utilities employ bespoke approaches to evaluating load additions, negotiating service agreements, and allocating upgrade costs. For developers, this fragmentation means timelines, processes, required studies, and cost responsibility vary widely across jurisdictions. The result is high search and negotiation costs, significant uncertainty about project viability, and wide variation in the effective “entry fee” for interconnecting to a transmission grid. In this context, interconnection standardization makes sense.

4. The policy question before FERC

The ANOPR explores whether FERC should establish a standardized national framework for interconnecting large loads to the bulk transmission system. One central question is whether such a framework would reduce barriers to efficient load integration, strengthen reliability, and promote nondiscriminatory access, or whether extending the existing generator-style interconnection regime to large loads would add complexity, reinforce outdated assumptions, and discourage the flexible, grid-supportive behaviors modern power systems require.

This comment evaluates that question through the lenses of transaction-cost economics, regulatory economics, and the operational realities facing utilities, grid operators, and large-load developers.

III. Proposition 1: Flexibility as the Central System Value

The ANOPR proposes expedited interconnection queue access for flexible large loads (Principle 7). Flexibility has become the binding constraint and the highest-value capability in modern power systems. Historically, both supply and demand have been price-inelastic. Supply is inelastic because new generation, transmission, and distribution infrastructure require long lead times, substantial capital, and multiple regulatory approvals. Demand for electricity is inelastic because electricity is an input with very few substitutes in nearly all economic activity, and because for data centers, the opportunity cost of curtailment is high due to the expected value of the computation. Demand for transmission is similarly inelastic due to the strict complementarity of the electricity transaction and the transmission transaction. AI inference and training workloads create large economic value per kilowatt-hour and cannot easily be deferred without eroding that value. When both sides of the market respond weakly to price, the outcome of demand growth is price increases.

In this environment, we should think of the opportunity for flexibility as the system’s “shadow price.” In constrained optimization, the shadow price measures the marginal value of relaxing a binding constraint. Here, the binding constraint is the inability of either supply or demand to adjust rapidly to other changes. Because supply cannot expand quickly and because demand cannot curtail without foregoing valuable activity, any resource or behavior that provides operational flexibility relaxes this constraint and thus has high marginal value. Flexibility restores the ability to adjust and to improve coordination.

1. Large load flexibility

Modern data centers possess far greater latent flexibility than legacy regulatory categories typically recognize, due in large part to recent and ongoing innovation. Advances in digital control systems, workload orchestration, and power-electronics architectures now permit these facilities to modulate demand across multiple time scales. Norris et al. (2025) quantifies this potential: it finds that, with modest curtailment of just 0.25 %–0.5 % of annual uptime, U.S. balancing authorities could integrate nearly 76 GW–98 GW of new large loads using existing capacity without disruptive grid upgrades.⁷

Four categories illustrate these emerging capabilities:

- **Operational flexibility through workload shifting.** Modern AI inference and training shapes are increasingly schedulable: inference tasks can shift across clusters and geographies, and training jobs can be staged to avoid peak-stress intervals. As distributed compute architectures and global traffic-balancing systems mature, the economic feasibility of temporal and spatial work shifting improves markedly, a finding echoed in the Norris et al.’s concept of “spatial flexibility” and “temporal flexibility”.
- **UPS- and storage-based fast response.** Uninterruptible power supply (UPS) systems originally designed for IT equipment protection can now function as fast frequency-response assets by modulating inverter-battery power flows on sub-second time scales. Norris et al. frames such on-site storage and generation as elements of “on-site power and storage” flexibility—capabilities that lower the incremental burden a new load places on the system.
- **Hyper flexible loads.** Flexible, and particularly hyper-flexible loads such as new technologies like thermal energy storage (TES), represent a significant opportunity for enhancing market performance. Long duration, highly flexible storage technologies with high charge to discharge ratios can provide continuous energy output over multi-day periods while only charging during periods of low energy demand and prices, when there is typically minimal stress on the grid. This operational profile allows these hyper flexible loads to avoid periods of peak scarcity while absorbing energy during low-value hours. In doing so, these hyper flexible and price sensitive loads improve price formation, mitigate congestion, reduce wasted energy from existing generators, and improve utilization of existing infrastructure. Hyper flexible loads like TES are being installed now with the operational capability and business model incentive to make robust, enforceable and dependable flexibility commitments. These loads have load factors of approximately 33% and can ramp up and down within milliseconds based on

⁷ Tyler H. Norris, Tim Profeta, Dalia Patino-Echeverri, and Adam Cowie-Haskell, [*Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in the U.S. Power System*](#). Nicholas Institute for Energy, Environment, and Sustainability, Duke University, 2025. Accessed November 21, 2025.

price or grid operator signals. Such commitments are fundamentally different in nature from commitments to curtail during emergencies made by high load factor customers with a high value of lost load, where such commitments are financial in nature and not directly enforceable by the grid operator.

- **Hybrid configurations with on-site generation.** Many developers are exploring on-site natural-gas generation, advanced fuel cells, and prospective small modular reactor (SMR) designs. Microgrid-ready architectures that incorporate a grid switch that enables islanded operation pending full interconnection offer dispatchable power, voltage support, and black-start capability. Norris et al. (2025) explicitly distinguishes loads with co-located generation as more flexible and therefore less likely to trigger transmission or generation capacity expansions.

Together, these capabilities transform large loads from passive grid takers into potential active system participants—able to provide ancillary services, reduce net peak demand, and mitigate the stability risks that reliability authorities increasingly identify. Notably, the Duke study defines “curtailment-enabled headroom” as a measure of how much additional load the system can absorb if flexible loads are available, and its results provide empirical weight for the argument that this capability is available now.

2. Current rules treat large loads as inflexible “problems”

Despite these emerging capabilities, the regulatory treatment of demand continues to assume that large loads are static, inflexible, and uncontrollable. This assumption permeates the interconnection and cost-recovery framework. Interconnection studies still model demand as a fixed block, ignoring the very types of flexibility that Norris et al. (2025) shows can reduce a load’s marginal impact on the system to a meaningful degree. Tariff structures often reinforce this static view, applying demand charges and cost-recovery riders that make no distinction between a load capable of sub-second frequency response and one that simply draws at its maximum nameplate level. The regulatory response to rising load forecasts has followed the same logic: delaying retirements of legacy fossil generators or extending the operating life of aging units, even where flexible loads could provide stability services that reduce these perceived needs.

This persistent mismatch between growing technological capability and backward-looking institutional assumptions weakens the alignment between private incentives and system needs. When flexibility is neither recognized nor rewarded, firms naturally invest in self-contained microgrids, behind-the-meter generation, and resilience-enhancing infrastructure that provide private reliability but contribute little to system-wide efficiency or stability. The result is a pattern of rational bypass driven not by a lack of technical solutions, but by regulatory structures that continue to treat demand as an inert quantity rather than a grid-supportive resource.

Flexibility also reduces the marginal cost of integrating large loads. A flexible load that can shift or reduce demand during peak conditions lowers the system's incremental capacity needs. When it co-locates with dispatchable on-site generation, it moderates congestion and reduces required network upgrades. Flexibility, therefore, improves reliability and helps bridge the pacing mismatch between data center speed-to-power and slower infrastructure expansion.

3. Implications for the ANOPR

The ANOPR rightly focuses on nondiscriminatory access and timely interconnection, but this framing risks overlooking the economic and reliability benefits of flexible demand. Its expedited queue access proposal is one step in the direction of valuing flexibility as a capability. In addition, a well-designed federal framework should:

- **Incorporate load-provided system services** into interconnection options and cost-allocation structures.
- **Enable partial rights, curtailable service, and hybrid configurations** that recognize the full range of flexible behaviors.
- **Acknowledge that flexibility affects both the magnitude and marginal cost of network upgrades.**

If implemented well, a federal framework could harness latent flexibility to improve system stability, reduce upgrade costs, and moderate the pacing mismatch between load development and infrastructure expansion.

If the regulatory regime treats all loads as inflexible and relies on embedded-cost allocation methods disconnected from marginal impacts, developers will rationally shift toward behind-the-meter solutions. That pattern is already visible in response to unreliable interconnection timelines and unpredictable upgrade costs. When grid rules fail to reward system-beneficial flexibility, investment flows toward private resilience that does little to strengthen the wider grid.

In a system marked by rapid load growth, variable renewable generation, and increasing dependence on power electronics, flexibility is economically and operationally indispensable. The regulatory framework must reflect that reality rather than rely on legacy assumptions that treat large loads as rigid and must-serve. The ANOPR's framework acknowledges the potential role of flexible loads, but could recognize the unique operational capabilities of various manifestations of flexible and hyper-flexible loads more explicitly as valuable resources for modernizing load interconnection. A federal regime that fails to integrate flexibility into interconnection and cost allocation will undervalue the very capabilities that matter most for system stability.

IV. Proposition 2: Legacy Interconnection Rules Create Significant Market Barriers

The second proposition is that the current interconnection framework is misaligned with the economic characteristics of modern large loads. It creates significant market barriers and raises transaction costs. These frictions are best understood through transaction-cost economics, especially Williamson's emphasis on asset specificity.⁸ Large-load interconnection requires grid infrastructure that is sunk, location-dependent, and ill-suited to alternative uses. At the same time, data center developers make their own site-specific investments—land, cooling systems, fiber connectivity, and construction—long before the grid is prepared to serve the facility. Both parties therefore face the classic hold-up risk: once one side commits irreversible capital, the other may act opportunistically.

From a competition perspective, interconnection rules play a central role in market entry conditions. When tariffs treat all loads as inflexible peak contributors, they distort competition, raise barriers to entry, and suppress innovation in flexible industrial technologies. The ANOPR's call for a modernized, consistent load-interconnection structure is therefore essential not only for reliability but for strengthening competition. By enabling flexible loads to compete on the basis of their actual operational profiles, the Commission would reduce regulatory frictions, support entrepreneurial risk-taking, and improve market efficiency. ERCOT has demonstrated the value of establishing rules that allow for integrated market participation of loads and hybrid facilities. Texas' Private Use Network rules, enabling participation of "energy parks" that may integrate load, storage, and generation components, has contributed to the region being one of the most competitive locations for new load and hybrid facilities

A durable standardized large load interconnection framework must begin with a clear understanding of the frictions that impede coordination among large-load developers, utilities, and grid operators. Transaction-cost economics offers a useful lens because it focuses on the features of the transaction—asset specificity, uncertainty, and frequency—and identifies the institutional arrangements most capable of reducing those costs. Large-load interconnection displays all three characteristics. It involves highly specific assets with limited alternative uses, it must navigate deep uncertainty surrounding load development and upgrade timelines, and it requires repeated interactions among parties with imperfectly aligned incentives.

1. Large-load interconnection as a high-asset-specificity transaction

Transmission upgrades needed to serve large loads are sunk investments that exist primarily, but often not exclusively, to serve the requesting facility. Similarly, the load itself often reflects investments tailored to a specific site that cannot easily relocate without forfeiting substantial value. This symmetry of sunk, site-specific investment generates two forms of hold-up risk. Utilities worry that if a developer cancels or delays a project, ratepayers will be saddled with

⁸ Oliver E. Williamson, "[Transaction Cost Economics](#)," *Handbook of Industrial Organization*. Elsevier, 1989, pp. 135-182.

stranded assets. Developers, having acquired land and begun construction, fear that an incumbent transmission owner may delay studies, escalate costs, or impose onerous conditions.

The hybrid or co-location contracts that large loads and generators are entering are system-enhancing examples of the “make or buy” decision in institutional economics.⁹ The decision to interconnect (Buy) or co-locate/build on-site generation (Make) is determined by the relative transaction costs of each path. Long-term contracting through co-location is often a rational economic response to high transaction costs and reliability risks in the market, given asset specificity and the opportunity for bilateral hold-up. Co-location is a market response to the pacing mismatch between data center speed-to-power and infrastructure construction timeframes.

The risks of asset specificity and holdout are magnified by the sheer scale of today’s loads. A facility that once required 20 MW may now request 500 to 1,000 MW at a single location, raising the stakes of interconnection negotiations and amplifying the value of credible commitments, transparent processes, and predictable cost-allocation rules. Where these features are absent, both utilities and developers hedge against risk, creating a self-reinforcing cycle of caution and delay. Ronald Coase’s insight that scarcity is rationed through non-price mechanisms when prices fail to reflect marginal cost is directly relevant: the interconnection queue becomes a rationing device rather than a coordination tool.¹⁰

2. Legacy interconnection rules intensify these frictions

The existing interconnection framework rests on assumptions about load behavior that no longer reflect technological reality. It treats loads as passive and uncontrollable, despite the flexibility described earlier. Studies model load at its maximum nameplate demand, discounting the ability of flexible facilities to maintain power quality through on-site generation, UPS response, or automated load controls. Cost-allocation methods presume that upgrades are fully attributable to the requesting load, even when they also relieve preexisting constraints or support future growth.

Applying a generator-style interconnection regime to large loads risks entrenching these problems further. The generator process is designed for resources that inject power, not for large, flexible loads whose operational behaviors differ fundamentally from generators. Importing this regime without careful consideration of the other market rules and study procedures involved risks embedding outdated assumptions about load rigidity and peak coincidence, and fails to recognize the operational capabilities large loads can bring to system stability.

⁹ Peter G. Klein, “[The Make-or-Buy Decision: Lessons from Empirical Studies](#),” in Claude Menard and Mary Shirley, eds., *Handbook of New Institutional Economics*, Springer, Boston, MA 2005.

¹⁰ Ronald H. Coase, “The Marginal Cost Controversy,” *Economica* 13(51) (1946): 169-182.

3. Beyond interconnection: Market rules as barriers to flexibility

Interconnection is not the only institutional bottleneck confronting large flexible loads. Even when a resource can technically interconnect and is willing to provide grid services, legacy RTO and ISO participation rules can prevent it from doing so. The market categories, performance requirements, telemetry obligations, and settlement structures embedded in wholesale power markets were developed for a system dominated by conventional generators and static loads. As a result, many flexible resources, particularly those capable of fast response, hybrid operation, or multi-service functionality, struggle to participate on equal terms with incumbent technologies.

The experience of thermal energy storage (TES) illustrates the problem. FERC's current jurisdictional framework treats TES inconsistently depending on which energy vector it dispatches. A TES resource that charges with electricity and discharges only electricity is treated as a wholesale participant. But a TES resource that charges with electricity and discharges thermal energy to an industrial host becomes simultaneously a wholesale and retail customer, even though its grid-facing behavior is indistinguishable from a purely electrical TES asset. If the TES resource dispatches only thermal output, it is treated exclusively as a retail customer. These three configurations impose similar net loads and provide the same demand-valley-filling services, yet the regulatory treatment diverges sharply. The resulting complexity raises transaction costs, introduces legal risk, and inhibits investment in a technology that could improve system flexibility materially. Clarifying that wholesale purchases of electricity for conversion into thermal energy qualify as wholesale transactions would reduce barriers to TES participation, improve competitive neutrality, and support innovation in industrial energy management.

This regulatory fragmentation is not limited to TES. Many innovative load-side and hybrid technologies face similar structural barriers rooted in outdated market definitions. Hybrid resources as defined in the ANOPR often do not fit neatly into either generator or load categories, pushing developers into duplicative or incompatible participation pathways. Demand-side flexibility resources that can respond with granular, automated precision encounter telemetry, metering, and aggregation requirements designed for large generators rather than for distributed or fast-acting loads. Fast frequency response and inverter-based services, including those provided by UPS systems and behind-the-meter batteries, frequently cannot qualify for compensation because existing product definitions overlook the temporal precision and performance characteristics modern power-electronics technologies can deliver. Even aggregated flexible loads, from industrial process controls to AI workload shifting, remain ineligible to participate as unified market entities despite offering aggregate capabilities that rival or exceed those of utility-scale assets. In each of these cases, the barrier is institutional rather than technical. Market rules built to maintain reliability within a twentieth-century generation mix now constrain the participation of twenty-first-century flexible resources,

creating a growing misalignment between the system's emerging needs and the regulatory structures designed to procure the services that flexibility can provide.

From the perspective of institutional and transaction-cost economics, these market barriers resemble the same pattern observed in interconnection: when legacy rules impose high administrative burdens, misclassify flexible behaviors, or treat new technologies as edge cases, the system increases the cost of innovation. Flexible resources that could contribute to frequency control, peak reduction, congestion relief, or ramping support are instead assigned to categories that view them as deviations from normal operations. The outcome is predictable: firms invest in privately beneficial but system-inefficient solutions because the wholesale market rules do not provide a viable channel for monetizing their flexibility.

Lowering these barriers requires modernizing market participation categories, redefining ancillary service products to reflect high-resolution flexibility, and explicitly integrating load-modulation technologies into wholesale markets. Doing so would align market design with the system's evolving physical characteristics and enable flexible demand-side resources to participate on equal footing with traditional supply-side assets.

V. Proposition 3: Cost Allocation Reform Is Foundational

The third proposition is that cost allocation reform is foundational to efficient large-load integration. The ANOPR suggests that large loads and hybrids should bear the full cost of any network upgrades identified in the interconnection study process. The implicit reasoning is straightforward: if the load triggers the need for upgrades, the load should pay. At first glance, this approach appears consistent with the principle of cost causation. While framed as a neutral application of cost causation, it rests on an economically fragile premise: that the method producing upgrade assignments accurately reflects long-run marginal cost.

But this logic holds only if the assigned costs reflect the marginal cost of serving the load. In practice, most upgrade assignments do not. Embedded-cost and fully distributed cost (FDC) methods routinely generate distorted outcomes that shape siting choices, suppress flexibility, and push developers toward bypass strategies that undermine reliability.

Transmission upgrade cost assignment remains one of the least transparent components of the interconnection process. Because most jurisdictions rely on embedded-cost or FDC methods, the relationship between a project's marginal impact and its assigned upgrade cost is weak. Two similarly sized projects can face orders-of-magnitude differences in upgrade charges due to queue position or earlier network investments. These inconsistencies are incompatible with the "just and reasonable" standard, create substantial barriers to rational siting, and encourage developers to hedge through speculative queue entries, parallel site evaluations, and behind-the-meter self-generation.

The economic analysis of joint product cost allocation is grounded in the regulatory economics literature. Coase's analysis of the marginal-cost controversy showed that marginal-cost pricing is essential for allocative efficiency but insufficient to recover the fixed costs characteristic of capital-intensive networks.¹¹ Ramsey's inverse-elasticity rule and Boiteux's formulation of peak-load pricing and multi-part tariffs clarified how to recover fixed costs with minimal distortion and how to reflect marginal system conditions in periods of scarcity.¹² These concepts underpin the cost-causation and beneficiary-pays principles of efficient rate design codified by Bonbright.¹³

1. The limits of embedded-cost and FDC methods

Embedded-cost and FDC methods allocate fixed and sunk transmission costs using historical accounting data and administrative formulas about contribution to peak load, usage shares, or historical averages, rather than actual marginal impacts. This disconnect produces several predictable distortions:

- **Non-linearity:** Costs change discontinuously depending on queue position or the triggering of discrete investments.
- **Lumpiness:** A single entrant may be assigned the full cost of an upgrade sized to serve future growth.
- **Backward-looking allocation:** Assignments embed historical planning decisions unrelated to current marginal needs.
- **Elasticity blindness:** All megawatts are treated alike even when some are highly flexible or deferrable.

Under conditions of rapid load growth, these weaknesses become more consequential. A 500–1,000 MW request can trigger substation expansions or major transmission additions, many of which are joint, lumpy investments intended to serve long-term development with multiple beneficiaries. Assigning the entire cost to the final entrant is not marginal cost causation; it is an artifact of administrative allocation.

Moreover, small differences in modeling assumptions—dispatch patterns, contingencies, topology—can yield substantially different upgrade requirements. A load may be assigned a major upgrade not because it caused the need but because it happened to be evaluated last. That outcome highlights why upgrade cost “stickiness” persists even after Order No. 2023 reforms.

¹¹ Ronald H. Coase, “[The Marginal Cost Controversy](#),” *Economica* 13(51) (1946): 169-182.

¹² Frank P. Ramsey, “[A Contribution to the Theory of Taxation](#),” *Economic Journal* 37(145) (1927): 47-61; Marcel Boiteux, “[Peak-Load Pricing](#),” *Journal of Business* 33(2) (1960): 157–79; Marcel Boiteux, “[On the Management of Public Monopolies Subject to Budgetary Constraints](#),” *Journal of Economic Theory* 3(3) (1971): 219-240. See also Ronald R. Braeutigam, “[Optimal Policies for Natural Monopolies](#),” *Handbook of Industrial Organization* Volume 2. Amsterdam: Elsevier, 1989, pp. 1289-1346.

¹³ James C. Bonbright, *Principles of Public Utility Rates*. Columbia University Press, 1961.

FDC methods are inherently arbitrary and detach cost allocation from marginal impact.¹⁴ As Bonbright emphasized, these approaches rely on accounting conventions rather than economic causation, producing allocations that bear little relationship to the incremental costs imposed by individual users. Braeutigam further showed that such methods suppress efficient price signals in network industries by forcing fixed costs into volumetric or peak-based charges that distort behavior. The result is a framework that obscures the economic rationale for efficient network use and encourages siting, consumption, and investment decisions at odds with system needs.

The “100 percent upgrade cost” principle is only sound if the underlying methodology is economically sound. It is not.

2. Ramsey-Boiteux as the correct benchmark

Rather than relying on embedded-cost or fully distributed cost formulas, the Ramsey–Boiteux framework offers the standard economic approach for recovering fixed network costs with minimal distortion.¹⁵ In natural monopoly settings, where marginal-cost pricing cannot cover the full cost of lumpy, capital-intensive infrastructure, the Ramsey–Boiteux solution identifies how prices should deviate from marginal cost to preserve efficient behavior. Its central insight is simple: when fixed costs must be recovered through markups, those markups should be inversely related to each customer class’s elasticity of demand. Customers with inelastic demand can bear somewhat higher markups without inefficiently reducing use, while customers with elastic demand will cut consumption, relocate, or bypass the system if charged too much.

In transmission planning, most upgrades create joint, shared benefits across multiple customer types—retail customers, generators, flexible loads, and hybrid resources—each with different elasticities. The Ramsey–Boiteux logic applies directly in this multi-beneficiary context: the fixed cost of a lumpy upgrade should be allocated in proportion to each beneficiary’s elasticity, not according to the timing of queue entry or arbitrary embedded-cost rules. Doing so minimizes the aggregate distortion created by fixed-cost recovery. Highly elastic users—those capable of relocating, deploying on-site generation, or strategically managing load—should face lower markups because pushing them away imposes a large welfare cost. More inelastic beneficiaries, whose consumption is less sensitive to price, can bear a larger share of fixed-cost recovery with fewer adverse system effects. The result is a cost-allocation structure that better reflects actual marginal impacts and preserves efficient usage and siting decisions among all parties.

¹⁴ James C. Bonbright, *Principles of Public Utility Rates*. Columbia University Press, 1961, Chapter XVIII; Ronald R. Braeutigam, “[Optimal Policies for Natural Monopolies](#),” *Handbook of Industrial Organization* Volume 2. Amsterdam: Elsevier, 1989, pp. 1313–1315.

¹⁵ Egbert Dierker, “[The Optimality of Boiteux-Ramsey Pricing](#),” *Econometrica* 59 (1) (1991): 99–121.

Applied to large-load interconnection, this principle has clear implications. Modern data centers and other flexible loads are among the most elastic customers on the system. They can shift locations, adopt hybrid on-site generation, or build microgrids if confronted with high upgrade charges. Assigning them the full cost of lumpy network upgrades—many of which provide long-run benefits to multiple users—maximizes distortion and encourages inefficient bypass, behaviors that the ANOPR seeks to avoid.

A Ramsey-aligned allocation would instead assign lower markups to highly elastic loads, especially those that provide measurable flexibility or operational benefits, while allocating a greater share of fixed-cost recovery to inelastic beneficiaries. This approach is forward-looking, efficiency-oriented, and well suited to a system experiencing rapid demand growth and substantial technological change.

3. Why excessive upgrade charges are counterproductive

When cost-allocation methods ignore elasticity and flexibility, predictable patterns emerge:

- Developers avoid high-cost jurisdictions regardless of system need.
- Projects migrate to regions with lower allocators, distorting regional development.
- Firms invest in behind-the-fence microgrids, often relying on natural gas or fuel cells, to secure faster time-to-power.
- Loads that could provide UPS-based frequency response or controllable demand instead isolate themselves electrically.

These responses work directly against nondiscriminatory access, reliability, and efficient integration of new load.

As load grows rapidly, the economics of these distortions become more visible. Many upgrades triggered in interconnection studies are joint, forward-looking investments. Assigning their full cost to a single entrant is economically unsound and ignores the shared nature of transmission infrastructure.

The ANOPR's suggestion of 100% participant funding (Principle 8) assumes that a new load is a static addition to a fixed system. In a dynamic grid, a flexible load provides system-wide benefits by increasing asset utilization. Assigning 100% of costs to an entrant who creates positive spillover effects violates the beneficiary-pays principle and discourages the very dynamic, decentralized participation that modern digital technologies enable.

4. Toward economically sound cost allocation

FERC should not adopt a blanket rule obligating large loads to bear 100 percent of assigned upgrade costs without first reforming the allocation methodology to reflect marginal cost principles. A coherent federal framework would:

- **Use two-part tariffs** separating marginal operating costs from fixed network cost recovery.
- **Rely on long-run marginal cost (LRMC)** for forward-looking upgrade assessments rather than historical accounting.
- **Incorporate locational differentiation** so signals reflect true incremental cost rather than arbitrary allocations.
- **Provide reduced upgrade burdens for loads with verifiable flexibility**, including UPS-based fast response, curtailable service, and hybrid on-site generation.
- **Minimize bypass incentives**, ensuring flexible, grid-supportive loads face a rational economic case for interconnection rather than isolation.

Cost allocation should reflect the principle that customers pay the costs they impose on the system, minus any benefits they provide. Flexible loads that avoid peak demand (including based on grid operator signals) do not cause peak-driven upgrades or capacity additions and should not be assigned such costs. When modeling assumptions reflect enforceable commitments and hybrid facility strategies, cost assignments become more accurate and more consistent with competitive neutrality. This change in methods, alongside a move away from FDC and toward Ramsey-Boiteux allocation methods, improves the efficiency of investment decisions and ensures that tariff design does not favor legacy technologies over innovative ones.

Aligning cost allocation with marginal impacts ensures that flexible loads remain part of the bulk system rather than retreating into private microgrids. It allows large loads to integrate as reliability-enhancing resources rather than becoming disconnected pockets of generation and demand. Most importantly, it supports the ANOPR's objective of enabling timely, efficient, and nondiscriminatory access to the transmission system.

VI. Design Principles Derived from Propositions 1–3

The analysis in Sections I–V points toward a coherent set of design principles for a modern federal approach to large-load interconnection. These principles flow directly from the economics of flexibility, the transaction-cost burdens of legacy rules, and the need for marginal-cost-aligned cost allocation and fixed cost recovery.

A sensible framework begins by treating flexibility as a system value rather than an operational irregularity. Modern large loads can supply flexibility that substitutes for missing supply elasticity and reduces marginal system costs. Regulatory structures must therefore embed their capabilities into interconnection studies, market-participation pathways, and cost allocation.

The federal role should also focus on reducing transaction costs. Many of the barriers examined earlier arise because interconnection and market rules treat loads as passive, uncontrollable entities, which increases uncertainty and magnifies hold-up risk. A

well-designed framework lowers these frictions and supports credible commitments that reflect the operational realities of flexible loads.

Finally, cost allocation must be anchored in marginal-cost principles and informed by the Ramsey–Boiteux logic for recovering fixed network costs. Transmission expansion is lumpy and jointly beneficial; allocating those fixed costs through embedded-cost formulas is economically incoherent and distorts investment and siting decisions. A modern approach must instead align charges with beneficiaries’ elasticities and marginal impacts.

Together, these principles outline a federal role that is modular and adaptive, one that encourages innovation rather than constraining it.

VII. Recommendations for a Federal Framework for Large-Load Interconnection

The design principles translate into a set of recommendations that modernize the federal framework in a way that aligns interconnection processes with the operational capabilities of large, flexible loads.

A modern framework requires interconnection options that explicitly recognize and value flexibility. Curtailable or limited-service arrangements give developers access to the grid at lower cost and faster timelines when they can accept conditional service. Loads capable of demonstrating enforceable flexibility—whether through UPS-based response, TES operation, hybrid generation, or workload shifting—should be eligible for accelerated study treatment. Interconnection analyses should reflect these capabilities by evaluating the actual operational profile of the load rather than assuming peak nameplate demand during scarcity events.

Market-participation pathways must be modernized as well. Participation categories, performance requirements, and telemetry obligations within RTOs and ISOs were designed for a system dominated by large generators and static loads. To enable modern demand-side flexibility, these rules must evolve. Flexible loads and hybrid resources need pathways that treat them as legitimate grid participants capable of supplying ancillary services, stability support, and demand response. Clarifying the jurisdictional status of technologies such as TES would resolve inconsistencies that currently prevent these resources from accessing wholesale markets on competitive terms.

The framework should also support competitive and transparent mechanisms that reduce strategic behavior and ease siting. Merchant and independent transmission projects can provide alternatives when incumbent processes lag. Tradable queue positions and upgrade rights can improve project sequencing and reduce incentives for speculative delay. Transparent hosting-capacity information and standardized modeling assumptions can lower search costs and provide developers with visibility into where their flexibility is most valuable and where it least strains the system.

VIII. Economically Sound Cost Allocation Framework

The analysis in Section V demonstrates that cost allocation is a pivotal element of an efficient interconnection regime. The inadequacies of embedded-cost and FDC methodologies are by now well understood. These approaches rely on historical accounting rather than economic causation, producing non-linear, backward-looking, and elasticity-blind assignments that distort siting decisions and create inconsistent outcomes across similar projects.

Embedded-cost allocation obscures marginal impacts, embeds sunk costs into pricing structures, and encourages consumption and investment patterns inconsistent with efficient network use.

Moving toward an economically coherent framework requires adopting the logic of Ramsey–Boiteux pricing for joint, lumpy transmission investments. The central insight is that fixed network costs should be recovered in a way that minimizes distortions, which means allocating the burden based on beneficiaries’ relative elasticities. In transmission planning, upgrades almost always benefit multiple user classes with differing degrees of elasticity, including retail customers, generators, flexible loads, and hybrid resources. A Ramsey-consistent approach allocates fixed costs across these classes in a way that imposes lower markups on elastic users—those capable of relocating, self-supplying, or modulating consumption—and higher markups on inelastic ones. This logic applies directly to modern large loads, which tend to be among the most elastic participants in the system. Charging them the full cost of lumpy upgrades predictably induces bypass, undermining both efficiency and reliability.

A forward-looking, marginal-cost-based approach therefore requires more than abandoning FDC. It calls for LRMC-based upgrade assessments, cost reductions for loads that provide verifiable flexibility or hybrid on-site generation, and tariff designs that separate marginal operating costs from fixed network recovery. These elements create a cost-allocation framework that reflects actual system impacts, minimizes bypass incentives, and encourages the integration of grid-supportive flexibility into the bulk system.

IX. How the Recommended Framework Advances the Objectives in the ANOPR

A federal framework built on these principles directly advances the ANOPR’s goals of timely, orderly, and nondiscriminatory interconnection. Flexibility-centered interconnection options and transparent study processes shorten timelines, reduce queue congestion, and improve alignment between load development and grid expansion. By standardizing modeling assumptions and requiring clearer documentation of study practices, the framework also increases predictability, reduces disputes, and improves coordination across regions.

Modernizing market rules and integrating flexible loads into ancillary service products promotes nondiscriminatory access. Treating flexible loads on equal terms with traditional resources recognizes their ability to contribute to frequency response, peak reduction, and congestion

management. These capabilities strengthen reliability at a time when the system is increasingly dependent on inverter-based resources and facing rapid load growth.

Finally, economically coherent cost allocation supports efficient siting, reduces unnecessary infrastructure investment, and minimizes the incentives that drive developers toward behind-the-fence microgrids. By aligning fixed-cost recovery with beneficiaries' elasticities and marginal impacts, the framework supports innovation in demand-modulating and hybrid resources while ensuring that cost responsibility remains consistent with economic principles.

X. Conclusion

The structural load growth associated with AI and data center expansion presents a real challenge, but one well within the capability of a modernized regulatory framework designed to recognize and reward flexibility. Legacy interconnection rules, market categories, and cost-allocation methodologies were developed for a system with very different characteristics. Left unchanged, they will suppress flexibility, slow innovation, and encourage inefficient bypass.

The analysis in this comment shows that a principles-based, modular federal approach—one that integrates flexibility into interconnection and market design, reduces transaction costs, and aligns cost allocation with marginal impacts—offers a more effective path forward. Such a framework will allow FERC and DOE to meet the moment: enabling timely, orderly, and nondiscriminatory integration of large loads while strengthening system reliability, improving market performance, and supporting the evolution of a flexible, resilient, and innovation-ready electricity system.

I respectfully request that the Commission take these initial comments into account as it considers the ANOPR and the development of a notice of proposed rulemaking in this proceeding.